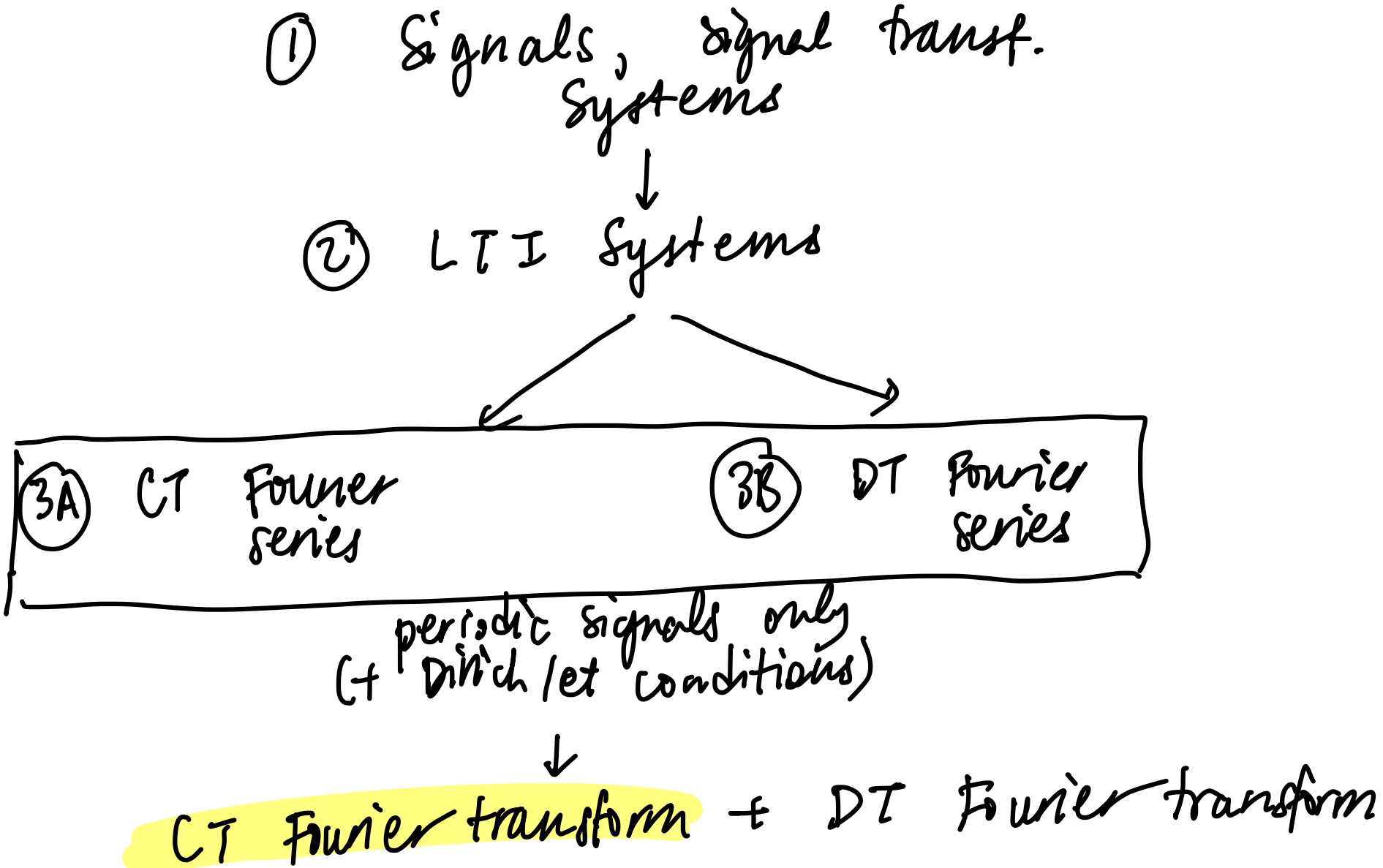
ELEC 221 Lecture 11
The CT Fourier Transform

Thursday 15 October 2026

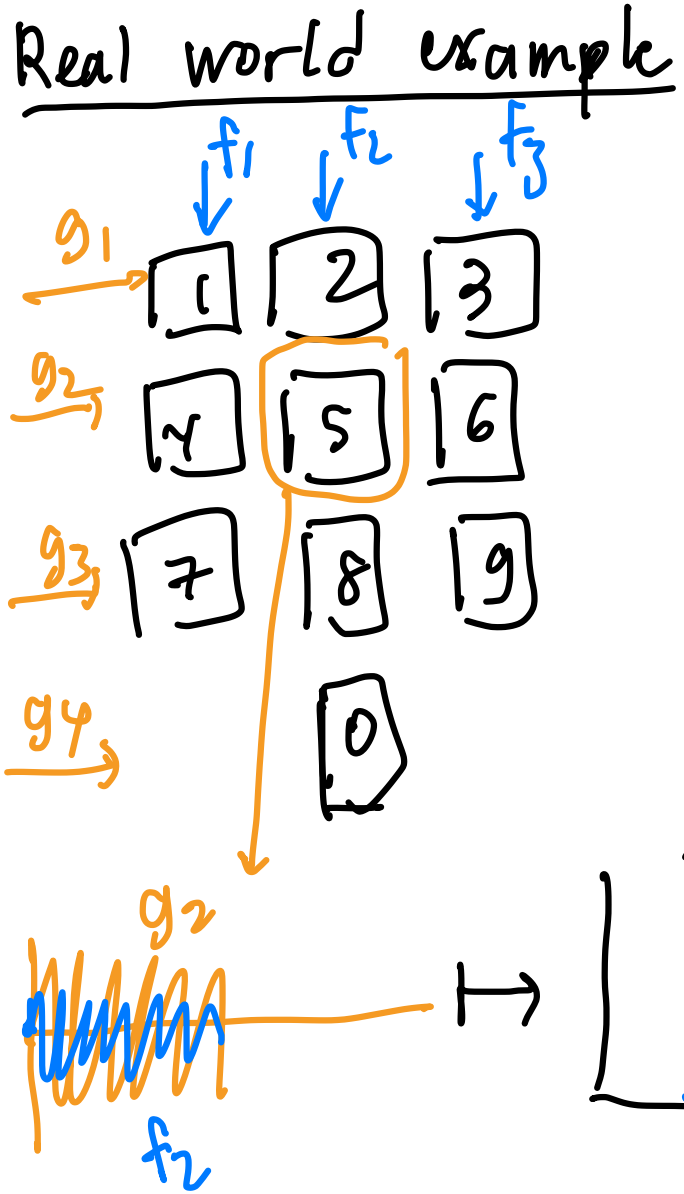
Announcements

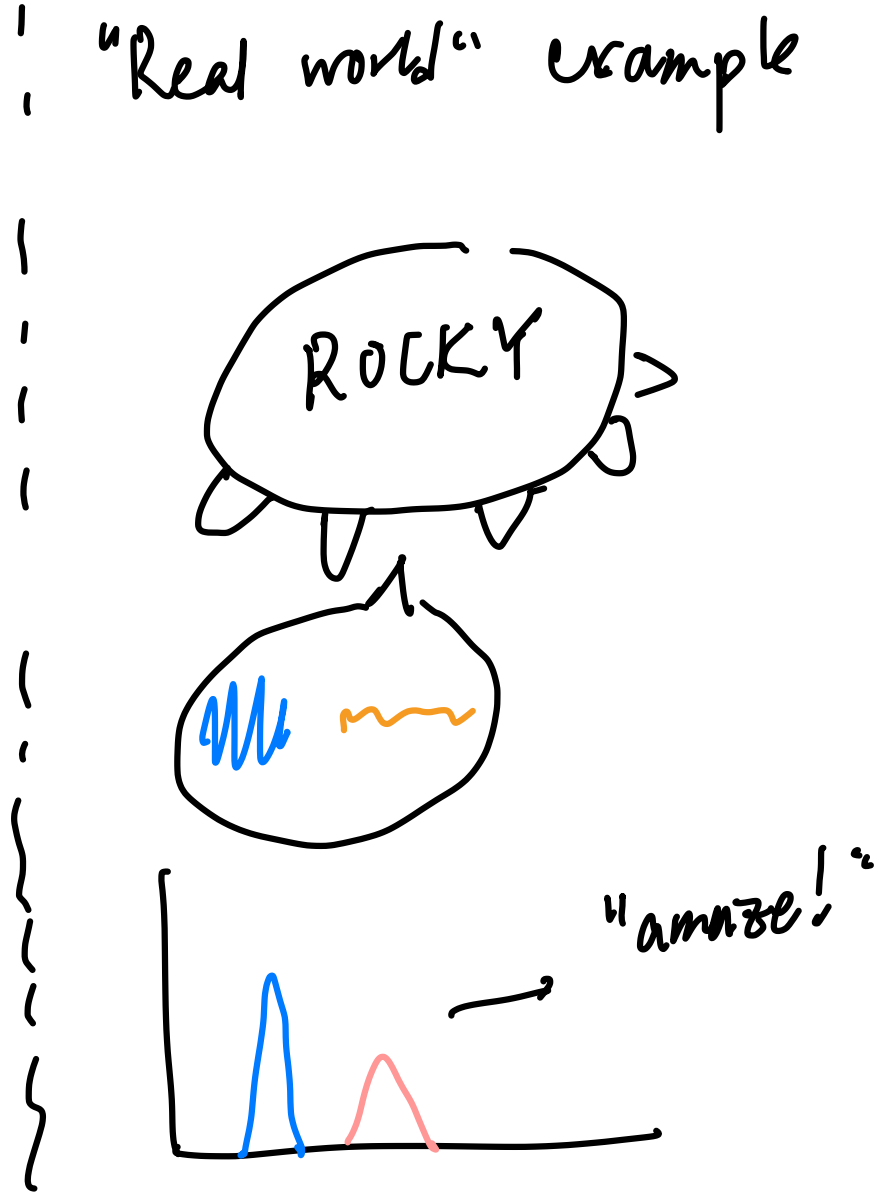
- Practice midterms available on PrairieLearn ~~by end-of-week~~ now.
- Quiz 6 Tuesday
- Tutorial assignment 3 on Monday (image processing)

Where are we going?



Where are we going?





Today

Learning outcomes:

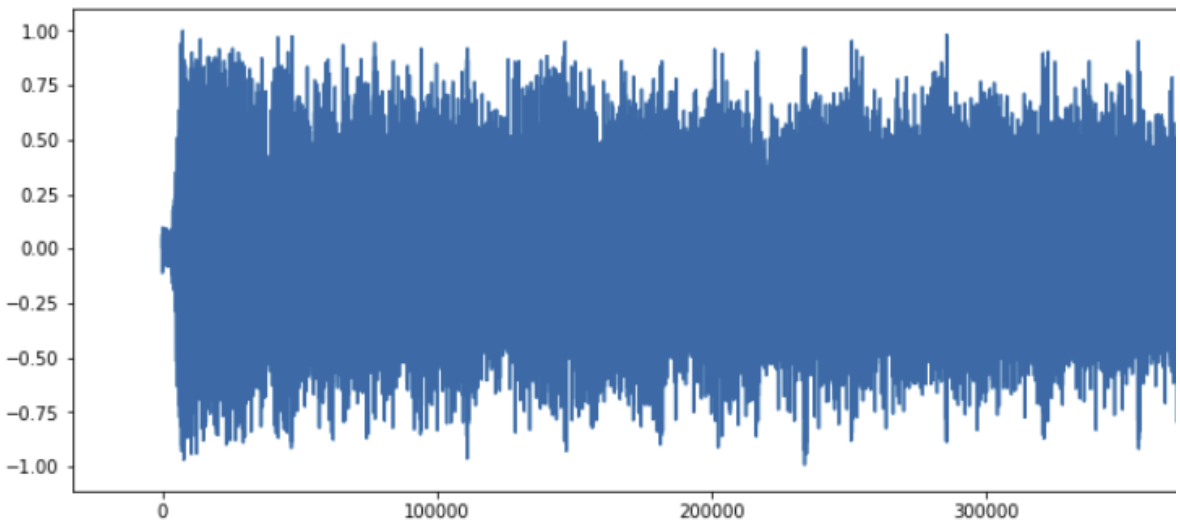
- Distinguish between the CT Fourier series and Fourier transform
- Define the CT Fourier transform and compute the Fourier spectrum of a CT signal
- State sufficient criteria for a signal to have a Fourier transform

Fourier series

In tutorial assignment 2, you worked with signals that *weren't periodic*:

```
plt.plot(audio)
```

```
[<matplotlib.lines.Line2D at 0x7f940880e370>]
```



But, we were still doing *something* with Fourier analysis:

```
fourier_spectrum = np.fft.rfft(audio)
```

The Fourier transform

The **Fourier transform** generalizes the Fourier series to **aperiodic signals**. It involves all possible frequencies.

Fourier series:

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega t} \qquad c_k = \frac{1}{T}\int_T x(t) e^{-jk\omega t}\, dt$$

ω fundamental frequency

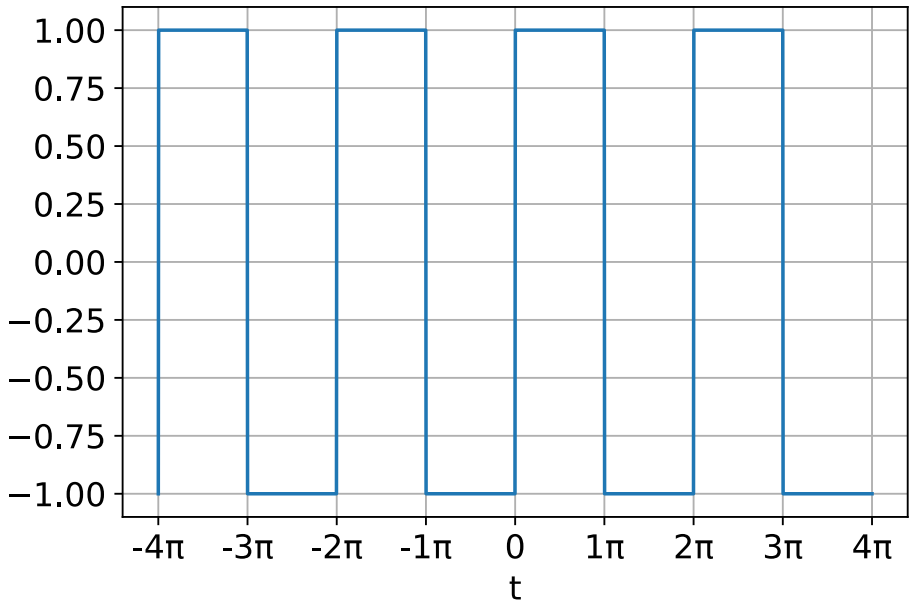
Fourier transform:

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega) e^{j\omega t}\, d\omega \qquad X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t}\, dt$$

How do we get here?

Towards the Fourier transform

Previously analyzed a 2π-periodic square wave:

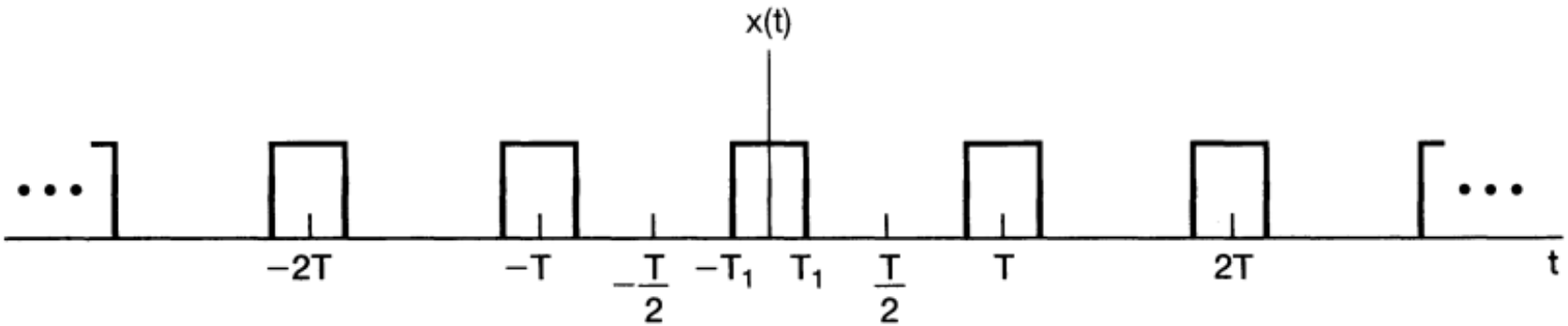


We derived its Fourier series representation

$$x(t) = \sum_{k=1}^{\infty} \frac{4}{k\pi} \sin(kt) \quad \text{only odd } k$$

Towards the Fourier transform

Let's generalize this. Consider the following square wave:

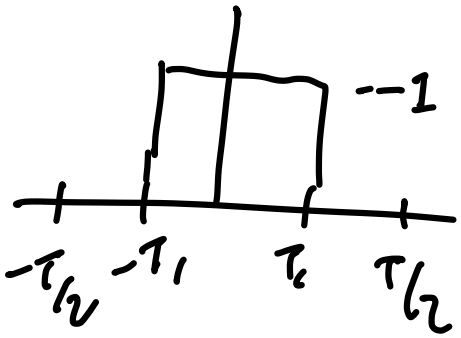


$$x(t) = \begin{cases} 1 & |t| < T_1 \\ 0 & T_1 < |t| < T/2 \end{cases}$$

Image credit: Oppenheim chapter 4.1

Towards the Fourier transform

$$x(t) = \begin{cases} 1, & |t| < T_1 \\ 0, & T_1 < |t| < T/2 \end{cases}$$



Compute its Fourier coefficients.

$$c_k = \frac{1}{T} \int_T x(t) e^{-jk\omega t}$$

Start with c_0:

$$c_0 = \frac{1}{T} \int_{-T/2}^{T/2} x(t)\, dt$$

$$= \frac{2T_1}{T}$$

Towards the Fourier transform

$$x(t) = \begin{cases} 1, & |t| < T_1 \\ 0, & T_1 < |t| < T/2 \end{cases}$$

Now the c_k:

$$c_k = \frac{1}{T}\int_{-T/2}^{T/2} x(t)\, e^{-jk\omega t}\, dt$$

$$= \frac{1}{T}\int_{-T_1}^{T_1} e^{-jk\omega t}\, dt$$

$$= -\frac{1}{jk\omega T}\, e^{-jk\omega t}\Big|_{-T_1}^{T_1}$$

$$= -\frac{1}{jk\omega T}\left[e^{-jk\omega T_1} - e^{jk\omega T_1}\right]$$

Euler: $e^{-jk\omega T_1} - e^{jk\omega T_1} = -2j\sin(k\omega T_1)$

$$= \frac{2}{k\omega T}\sin(k\omega T_1)$$

← or split $e^{-jk\omega t}$ into cos/sin and simplify based on symmetry or evenness/oddness.

Towards the Fourier transform

What does this function look like?

$$c_0 = \frac{2T_1}{T} \qquad c_k = \frac{2}{k\omega T}\sin(k\omega T_1)$$

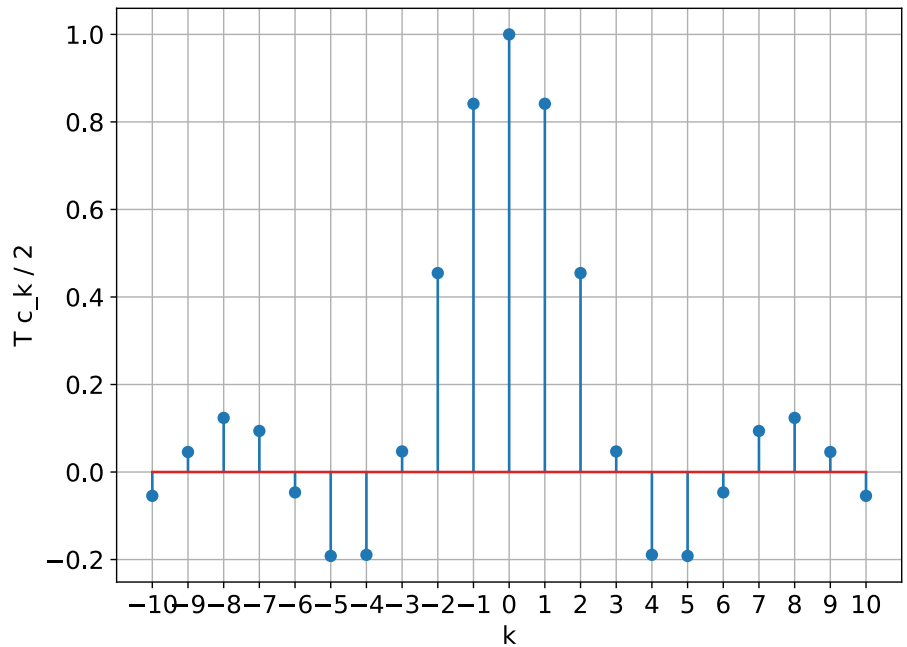
Rearrange and express as a function of k:

$$f(k) = \frac{Tc_k}{2} = \begin{cases} T_1 & k=0 \\ \dfrac{\sin(k\omega T_1)}{k\omega} & k\neq 0 \end{cases}$$

Let's plot this: set $T_1 = 1$, and $T = 2\pi$, so $\omega = \frac{2\pi}{T} = 1$.

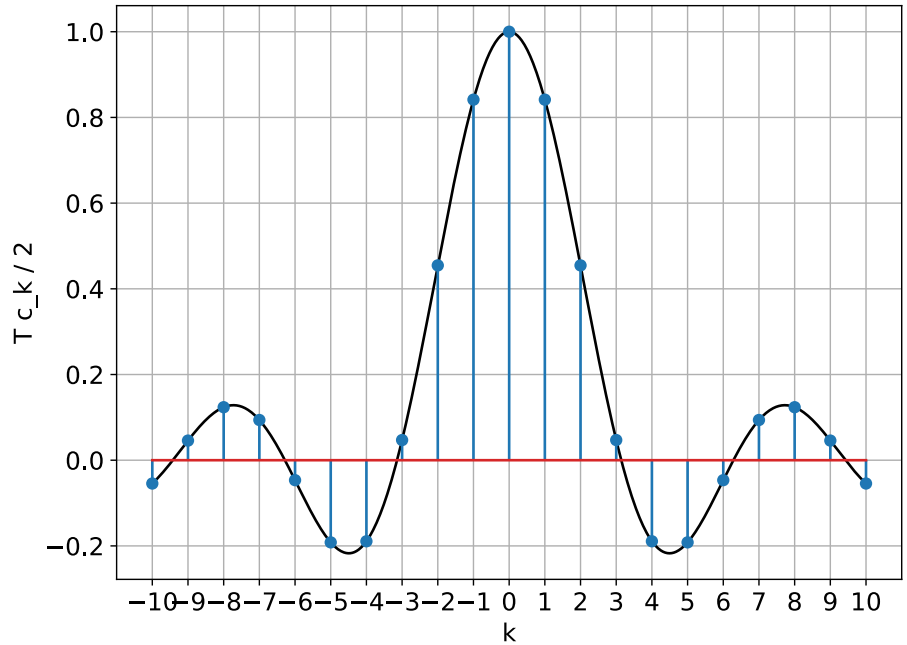
Towards the Fourier transform

$$f(k) = \frac{Tc_k}{2} = \begin{cases} T_1 & k=0 \\ \dfrac{\sin(k\omega T_1)}{k\omega} & k \neq 0 \end{cases}$$



$$T_1 = 1 \qquad T = 2\pi \;, \qquad \omega = \frac{2\pi}{T} = 1$$

Towards the Fourier transform



These are **samples** of the function

$$f(k) = \begin{cases} T_1 & k=0 \\ \dfrac{\sin(k\omega T_1)}{k\omega} & k \neq 0 \end{cases}$$

at *integer values of* k.

Towards the Fourier transform

Re-express this as a function of frequency, $\tilde{\omega} = k\omega$:

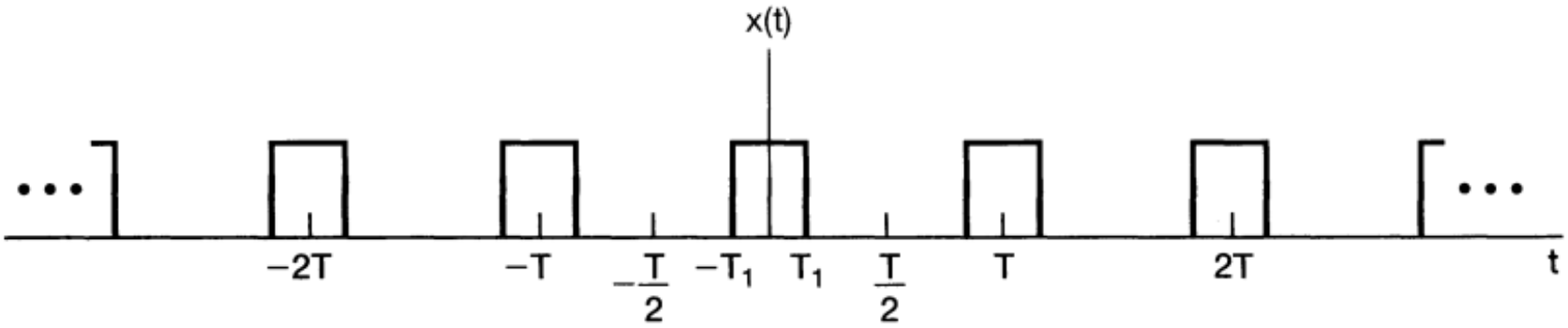
$$f(\tilde{\omega}) = \begin{cases} T_1 & \tilde{\omega} = 0 \\ \dfrac{\sin(\tilde{\omega} T_1)}{\tilde{\omega}} & \tilde{\omega} \neq 0 \end{cases}$$

The Fourier coefficients are samples of this function at *integer multiples* of fundamental frequency, $\tilde{\omega} = k\omega$, where $\omega = 2\pi/T$.

$$c_k = \frac{2}{T} f(\tilde{\omega}) = \frac{2}{T} f(k\omega)$$

Towards the Fourier transform

Suppose T grows, but T_1 stays the same:



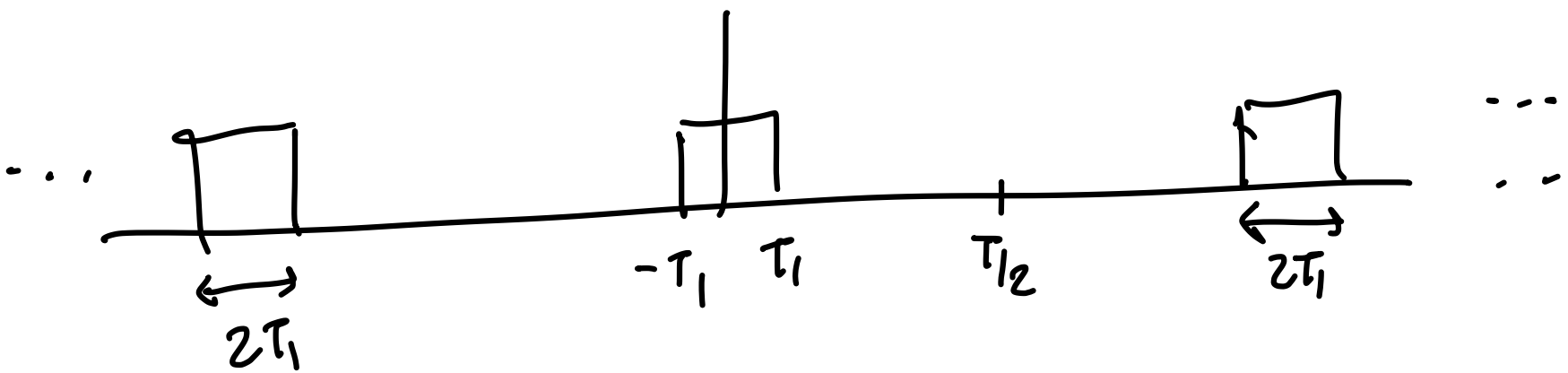


Image credit: Oppenheim chapter 4.1

Towards the Fourier transform

Initially the spacing of samples is integer multiples of $\omega = 2\pi/T$.

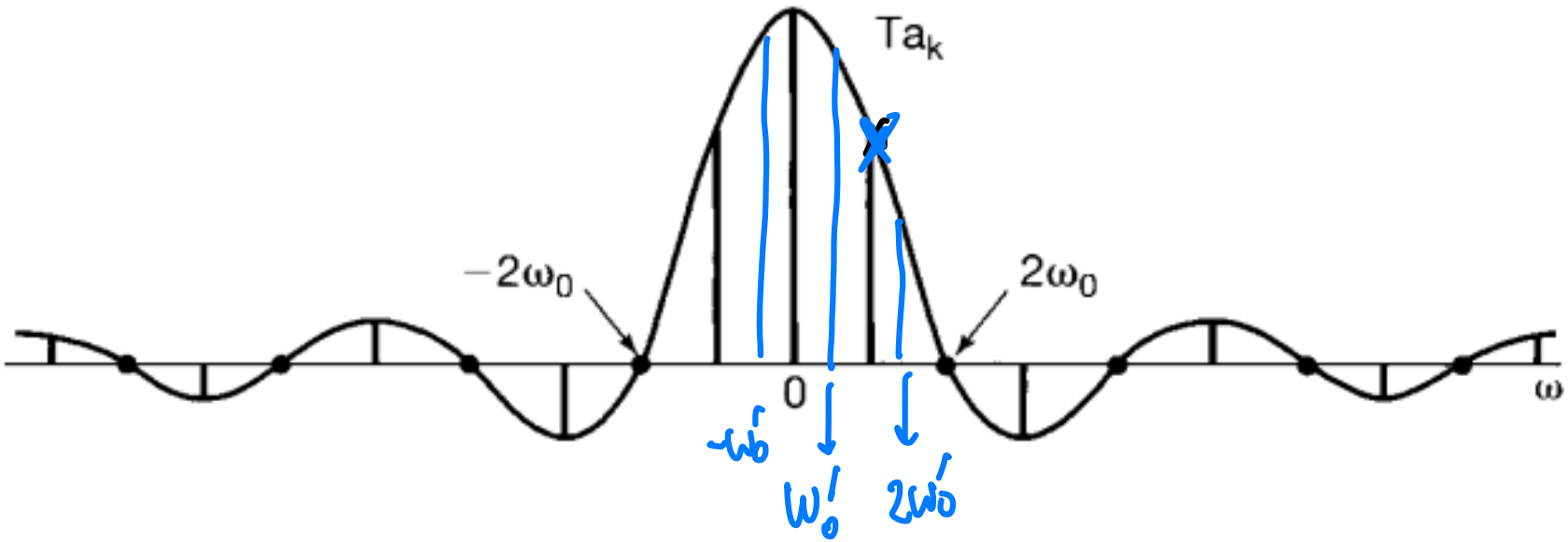
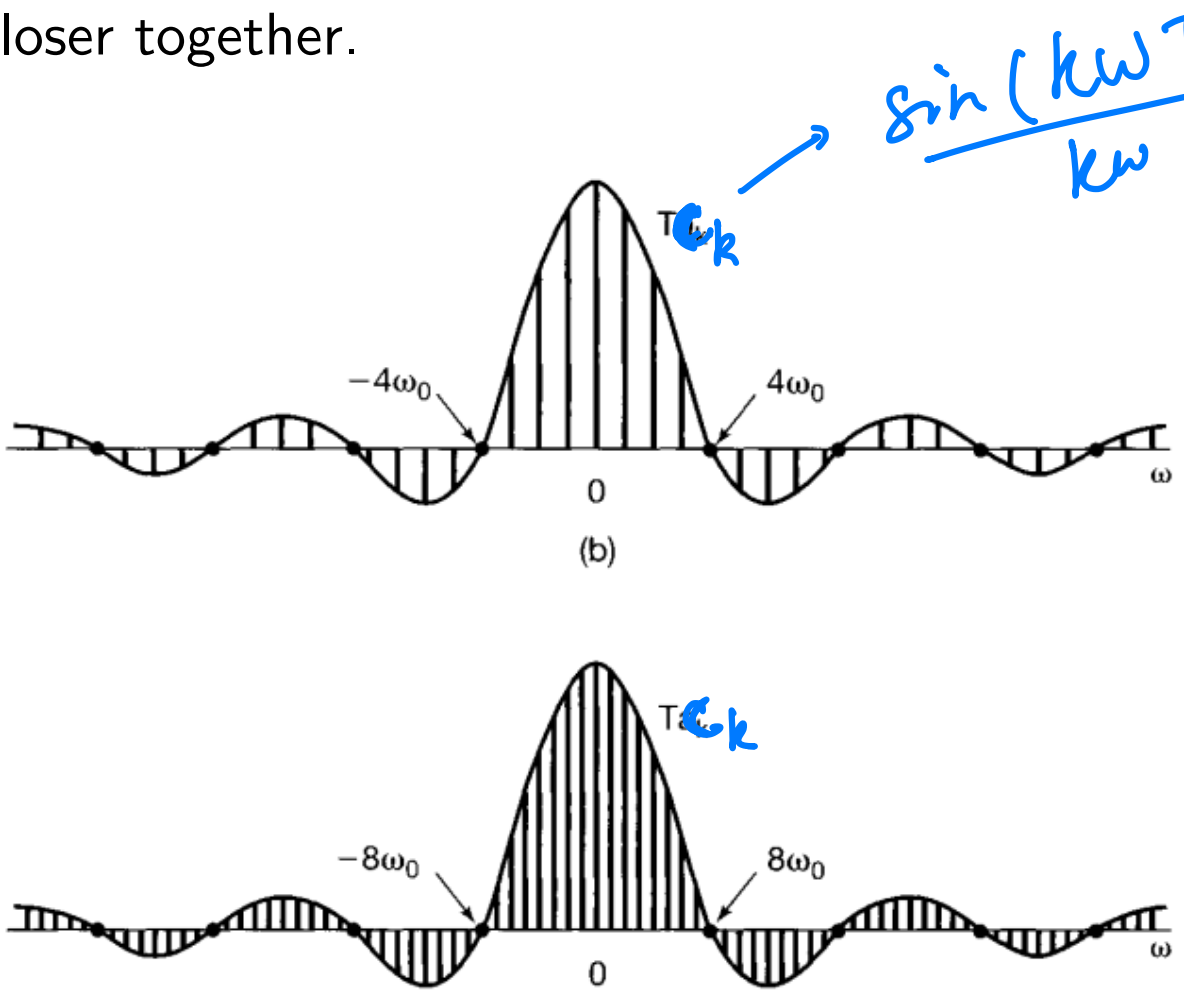


Image credit: Oppenheim chapter 4.1

Towards the Fourier transform

As T grows, $\omega = 2\pi/T$ becomes smaller so the integer multiples of it get closer together.



Towards the Fourier transform

Eventually, ω becomes so small. Instead of

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{-jk\omega t}$$

consider the sum as a continuous integral over ω:

$$x(t) \sim \int_{-\infty}^{\infty} c_\omega e^{j\omega t} d\omega$$

c_ω: some freq. dep. coefficient

How does this apply to aperiodic signals?

Towards the Fourier transform

Given any aperiodic signal $x(t)$, "pretend" it's periodic by constructing a **periodic extension**, $\tilde{x}(t)$ with period T.

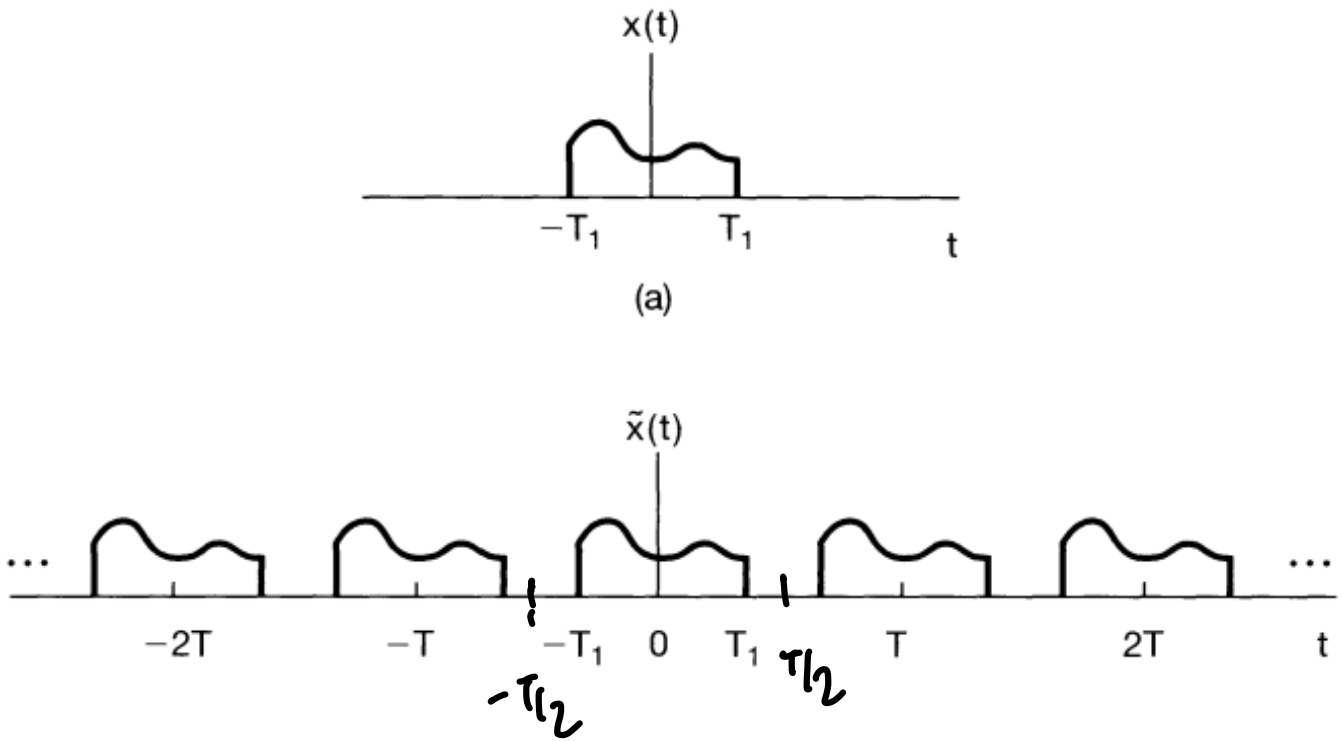


Image credit: Oppenheim chapter 4.1

Motivation: Fourier transform

We can express $\tilde{x}(t)$ as a Fourier series (where $\omega = 2\pi/T$):

$$\tilde{x}(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega t}$$

$$c_k = \frac{1}{T} \int_T \tilde{x}(t) e^{-jk\omega t} \, dt$$

Motivation: Fourier transform

What happens to the coefficients?

$$c_k = \frac{1}{T}\int_{-T/2}^{T/2} \tilde{x}(t)\, e^{-jk\omega t}\, dt$$

$$= \frac{1}{T}\int_{-T/2}^{T/2} x(t)\, e^{-jk\omega t}\, dt$$

$$= \frac{1}{T}\underbrace{\int_{-\infty}^{\infty} x(t)\, e^{-jk\omega t}\, dt}_{X(jk\omega)}$$

Let's define

so that

$$c_k = \frac{1}{T} X(jk\omega)$$

Motivation: Fourier transform

We can put this back in our Fourier series:

$$\tilde{x}(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega t}$$

$$= \sum_{k=-\infty}^{\infty} \frac{1}{T} X(jk\omega) e^{jk\omega t}$$

$$= \sum_{k=-\infty}^{\infty} \frac{\omega}{2\pi} X(jk\omega) e^{jk\omega t}$$

$$= \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} X(jk\omega) e^{jk\omega t} \omega$$

Motivation: Fourier transform

$$\tilde{x}(t) = \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} X(jk\omega)\, e^{jk\omega t}\, \omega$$

Consider what happens when $T \to \infty$...

1. $\tilde{x}(t)$ will look just like $x(t)$ for large enough T

$$\lim_{T\to\infty} \tilde{x}(t) = x(t)$$

2. ω will get smaller and smaller

$$\sum_{k=-\infty}^{\infty} X(jk\omega)\, e^{jk\omega t}\, \omega \to \int_{-\infty}^{\infty} X(j\omega)\, e^{j\omega t}\, d\omega$$

The Fourier transform

Inverse Fourier transform (synthesis equation):

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega)\, e^{j\omega t}\, d\omega$$

Fourier transform (analysis equation, or Fourier *spectrum*):

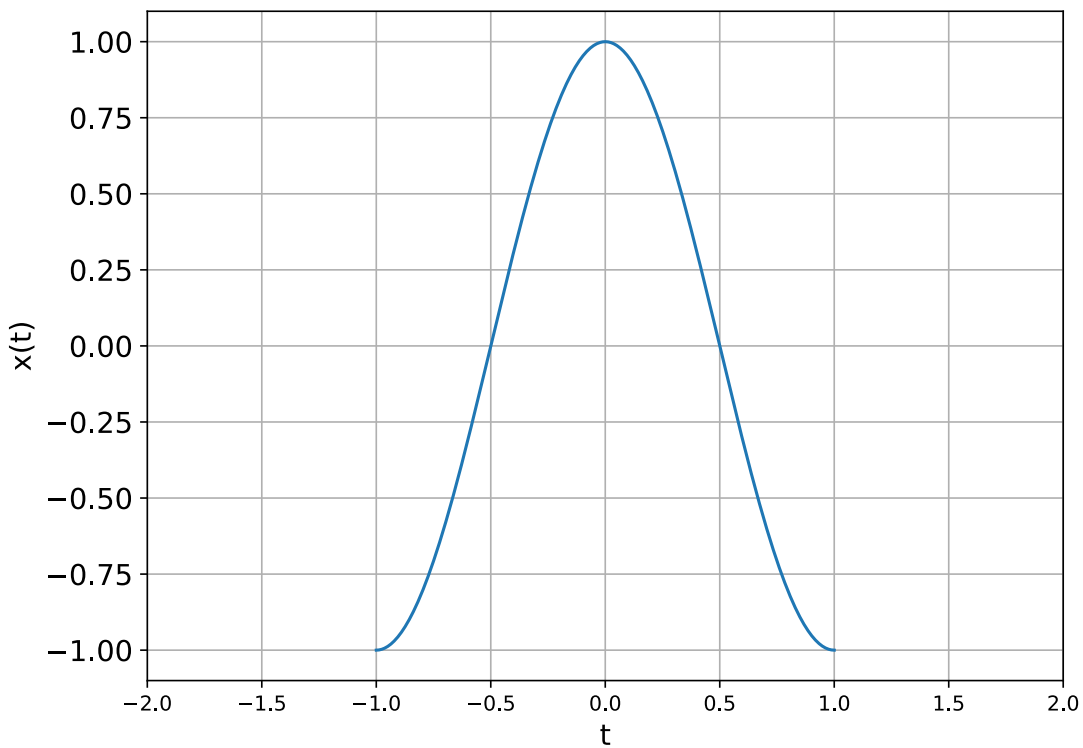
$$X(j\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt$$

Note: Sometimes the $1/2\pi$ prefactor appears on the spectrum, or sometimes both versions have $1/\sqrt{2\pi}$.

Example

Compute the Fourier spectrum of:

$$x(t) = \begin{cases} \cos(\pi t), & |t| \leq 1 \\ 0, & |t| > 1 \end{cases}$$



Example

$$x(t) = \begin{cases} \cos(\pi t), & |t| \le 1 \\ 0, & |t| > 1 \end{cases}$$

Start from the definition:

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt$$

$$= \int_{-1}^{1} \cos(\pi t)\, e^{-j\omega t}\, dt$$

apply Euler's formula

Example

$$X(j\omega) = \frac{1}{2}\int_{-1}^{1} e^{j(\pi-\omega)t}dt + \frac{1}{2}\int_{-1}^{1} e^{-j(\pi+\omega)t}dt$$

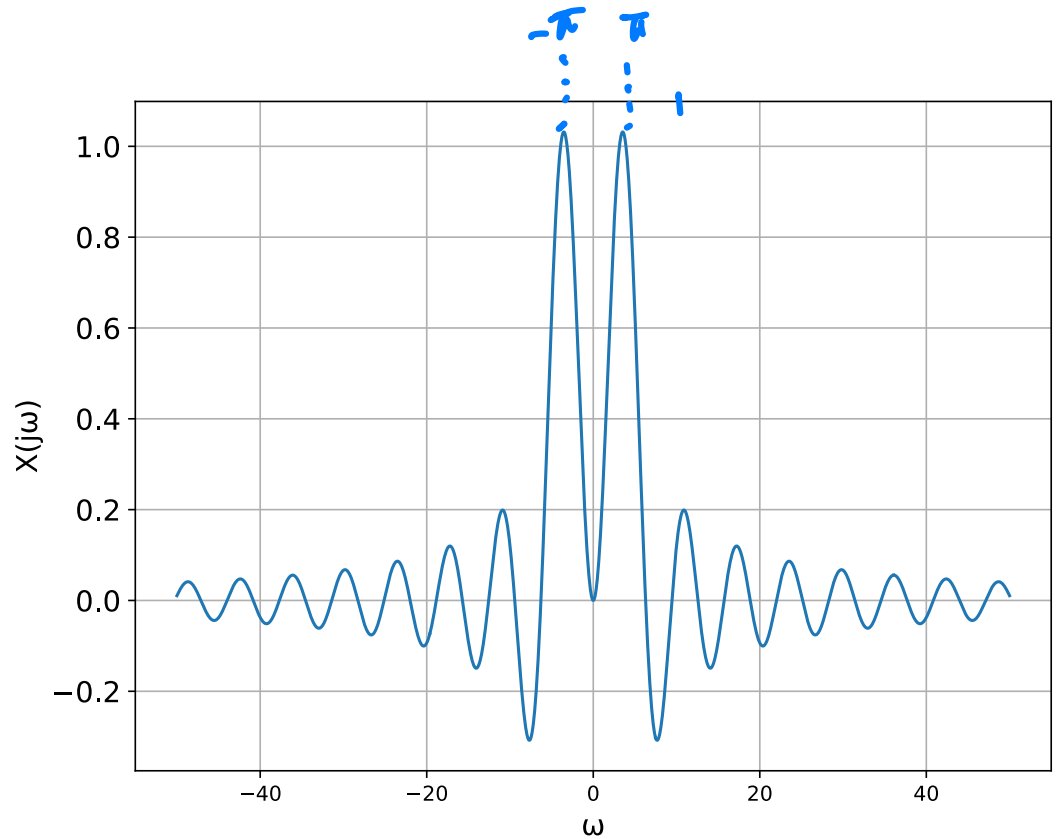
$$\vdots$$

$$= \frac{\sin(\omega)}{\pi-\omega} - \frac{\sin(\omega)}{\pi+\omega}$$

Example

$\cos(\pi t)$

$$X(j\omega) = \frac{\sin(\omega)}{\pi - \omega} - \frac{\sin(\omega)}{\pi + \omega}$$



For next time

Content:

- Dirichlet conditions for CT Fourier transform
- Convolution and multiplication property of the Fourier transform
- Time-frequency duality
- Differentiation and integration of FT

Recommended reading:

- From today's class: Oppenheim 4.0-4.2
- Suggested problems: 4.1-4.4, 4.6, 4.9, 4.21bcdgh, 4.22abde, 4.27
- For next class: Oppenheim 4.3-4.6